Simulating a CT department

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1 Introduction

In this report, we will simulate the current situation in a CT scan department. The simulation model will be used to determine certain performance indicators.

We will consider three types of patients that can arrive to the CT department:

- Emergency patients
- Inpatients
- Outpatients

Emergency patients have the highest priority, these patients arrive unexpectedly and will immediately be treated once they arrive.
Inpatients are patients that are already in the hospital. They also arrive unexpectedly, but can wait for some time before being treated. However, scanning them on the same day as they arrive is highly preferred.
Outpatients schedule their appointment at least the day before. These appointments are scheduled at 15-minute intervals. Every outpatient has a chance to not show up to their appointment.

In section 2, we describe the CT department and the performance measures we are interested in. Section 3 gives an overview of the simulation model and the assumptions we made. In section 4 we describe the data we collected during simulation, and how we use this data to determine confidence intervals for the performance measures we want. In section 5 we present our results and section 6 will contain the conclusions and recommendations.

2 Problem description

The CT department currently uses two CT scanners, both of which are available during office hours, i.e. between 8:00 and 16:00 on weekdays. Outside these office hours, one scanner will remain available for emergency patients and inpatients.
The department does not currently make use of a dynamic blueprint. Instead, they have the following policy for patient scheduling: outpatients are scheduled at the first free slot after they call to make an appointment (at the earliest the day after the call). Every morning (between 8:00 and 12:00) there are 4 slots per hour, in the afternoons (12:00 - 16:00) there are 3 per hour. If there are no free slots left in the week that they call, they will be put on a waiting list. Every Friday, all patients on this waiting list will be scheduled in the next week.
After their arrival, all three types of patients wait in the waiting room, which currently has three chairs. As soon as a scanner becomes free, the next patient will leave the waiting room in order to be treated. If there is an emergency patient waiting, that patient will always be the next to be treated. In- and outpatients are treated in order of arrival.

The performance measures we consider are:

- CT scanner utilization, both during and outside office hours
- Access times for outpatients (the number of days between the call and the appointment)
- Waiting time for emergency patients and outpatients (the time spent in the waiting room)
- Fraction of patients that have to wait outside the waiting room
- Fraction of inpatients with requests during office hours that cannot be treated during the office hours of that day.

3 Simulation model

3.1 Assumptions

In addition to the assumptions specified in the problem description (section 2), we made the following assumptions:

- The duration of a CT scan follows a Uniform(10,19) distribution.
- Emergency patients arrive according to a Poisson process with rate $\lambda_E = 24$ per day.
- Inpatient requests arrive according to the process specified in the project description.
- On average $\lambda_O = 23$ calls from outpatients arrive on a weekday during office hours according to a Poisson process.
- Each scheduled outpatient shows up with probability $p_s = 0.84$.
- The travel time for inpatients from the ward to the CT department follows a Uniform(9,15) distribution.

3.2 State variables

Our simulation keeps track of the following state variables:

- Number of patients currently being scanned
- Number of patients currently in the queue
- Number of inpatients waiting in the ward
- Number of outpatients on the waiting list

3.3 Events

The simulation uses six types of primary events, see also a flowchart in figure 1:

- Outpatient call
  When an outpatient calls, the patient is assigned to the next free slot. If no slot is available during the current week, the patient is added to the waiting list. After the call is processed, the next call is generated with the specified exponentially distributed inter-arrival time.
- Outpatient arrival
  Outpatients arrive at the time they are scheduled. If no scanner is currently available, the patient joins the waiting queue.
- Inpatient request
  When an inpatient scan request happens, the patient is placed on an inpatient waiting list, and the next inpatient requests is generated according to the specified Poisson process.

- Inpatient arrival
  After the CT department has called the ward and the inpatient has traveled to the department, the patient joins the queue, unless a scanner is already available, in which case the patient is treated immediately.

- Emergency patient arrival
  Once an emergency patient arrives, the patient joins the queue "at the front of the queue", because emergency patients always have priority over all other patients. If a scanner is already available the patient is scanned immediately. After the patient arrived, the next emergency patient arrival is generated according to the specified Poisson process.

- End of a scan
  When a CT scan is completed, the patient leaves the system, and the next person in the queue starts being scanned. If the queue is empty at that moment, the scanner becomes available for the next patient to arrive.

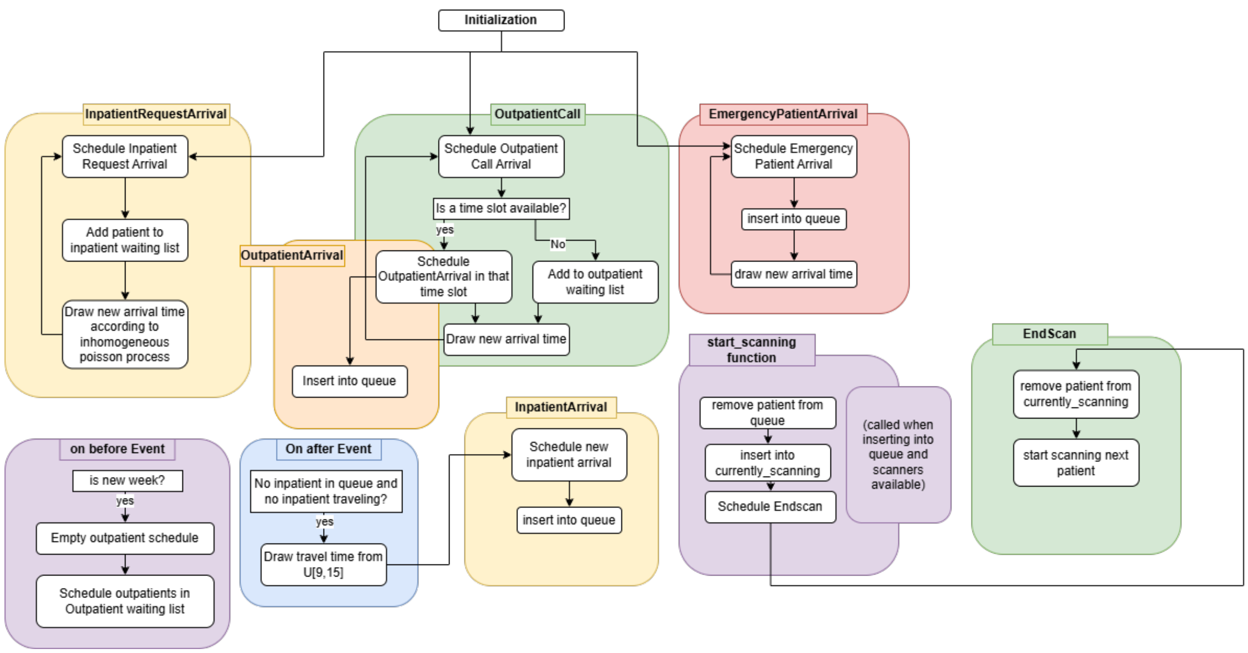

Figure 1: Flowchart of events

4 Statistical analysis

4.1 Batch method

We use the batch method to keep track of the needed statistics in our simulation. The regenerative method produced unstable estimates, because the states we chose as regeneration points, being an empty queue or an empty system, occurred very frequently, which meant we ended up with extremely short cycles.

4.1.1 Number of batches

We found that for all statistics we keep track of, with a precision of $\delta = 0.1$, the following expression holds for a number of batches $r = 50$:

$$r \geq \frac{t^2_{1-\alpha/2,r-1} S^2(r)}{\left(\frac{\delta}{1+\delta}\hat{X}\right)^2} \tag{1}$$

More specifically, the expression evaluates to the values in the following table:

Statistic	R.H.S. expression in equation (1)
Emergency patient waiting time	0.2462
Outpatient waiting time	0.7271
Total fraction of patients waiting outside	17.8651
Scanner utilization in office hours	0.2349
Scanner utilization outside office hours	0.0770
Access time outpatients	0.0682
Fraction inpatients called in office hours scanned outside office hours	8.6315

Table 1: Values of the right hand side expression in equation (1) for different statistics, when running without warmup period for a running time of 10^7 minutes.

4.1.2 Warm-up period

We found that a warm-up period of $2 \cdot 10^7$ minutes, or ~ 38 years, for a total running time of 10^8 minutes, or ~ 190 years the following expression was satisfied for all statistics:

$$\left| \frac{\frac{1}{2D}\sum_{k=1}^{2D} \bar{W}_{*,k}}{\frac{1}{D}\sum_{k=1}^{D} \bar{W}_{*,k}} - 1 \right| \leq 0.05 \tag{2}$$

More specifically, the left hand side expression of the above equation evaluates to the following for different statistics:

Statistic	L.H.S. expression in equation (2)
Emergency patient waiting time	0.0046
Outpatient waiting time	0.0059
Total fraction of patients waiting outside	0.0042
Scanner utilization in office hours	0.0010
Scanner utilization outside office hours	0.0006
Access time outpatients	0.0013
Fraction inpatients called in office hours scanned outside office hours	0.0256

Table 2: Values of the left hand side expression in equation (2) for different statistics with a warm up period of $D = 2 \cdot 10^7$ for a running time of 10^8 minutes.

A small note: for most of the statistics, a warm-up period of around $D = 10^6$ minutes was more than enough to satisfy the expression in equation (2), but the fraction of patients waiting outside and fraction of inpatients called in office hours scanned outside office hours required a way longer warm up period, because these events occur relatively infrequently. We chose not to ignore these statistics and increase the warm-up period and thus also the total running time, but this may not be necessary.

4.2 Verification cases

To test the validity of our model, we will reduce the system to a simpler system with an easy to compute value for some statistics of interest.

4.2.1 M/G/1 Queue

letting only emergency patients arrive and setting the total number of scanners available to 1 at any time, we reduce the system to an M/G/1 queue with arrival rate $\lambda = \frac{1}{60}$ per minute and service time $B \sim U[10, 19]$ minutes. This yields

$$W_q = 2.3846, \qquad L_q = 0.0397.$$

In our simulation, we find:

$$\text{Emergency patient waiting time} = 2.381, \qquad \text{Queue length} = 0.0398.$$

We also see that the scanner utilization in office hours 0.2218 and outside office hours 0.2451 are very close to the theoretical server utilization $\rho = \lambda \mathbb{E}[B] = \frac{14.5}{60} = 0.2417$.

4.2.2 M/M/1 Queue

We can further reduce the system to an M/M/1 queue if we let the scanning time be distributed like $B \sim \text{Exp}(\frac{1}{15})$. This yields

$$W_q = 5, \qquad L_q = \frac{1}{12} = 0.08333.$$

In our simulation we find:

$$\text{Emergency patient waiting time} = 4.9993, \qquad \text{Queue length} = 0.0834.$$

Again, we see that also the scanner utilization in office hours 0.2522 and outside office hours 0.2550 are very close to the server utilization $\rho = \frac{\lambda}{\mu} = \frac{15}{60} = 0.25$.

4.2.3 Verification cases conclusion

For both verification cases, the simulated values are very close to the theoretical values, the relative errors are below 1% for all of them, indicating that the logic has been implemented correctly in the simulation model.

5 Results

5.1 Analysis

We are asked if 2 CT scanners are enough to treat all patients. Note that during office hours there arrive on average 8 emergency patients, 23 outpatient calls and 21 inpatient requests. Since scheduling of outpatients and calling inpatients can be viewed as their own system (which should be stable, i.e. the queue lengths do not explode), where 21 inpatients that make a request go through the system during office hours and $23 \cdot p_s = 19.32$ outpatients, we can assume that, on average 48.32 patients need to be scanned during office hours. Scanning times are uniformly distributed on $[10, 19]$ minutes, meaning one scan takes on average 14.5 minutes. Thus, the two scanners will, on average, be busy for 5.8387 hours each during office hours on average. Then, each working day the scanner utilization should be 0.7298, meaning the overall scanner utilization should be 0.5213. Similarly, outside office hours there arrive on average 16 emergency patients, 0 outpatient calls and 6 inpatient requests, in total 22 patients. The one scanner is then, on average, busy for 5.3167 hours, meaning the scanner utilization outside office hours during working days should be 0.3323, meaning the total scanner utilization should be 0.3006. Since the scanner utilization fractions are strictly less then 1, we know that the system is stable, i.e. the queue size will not ‘explode’. This means that 2 scanners are enough for the hospital.

5.2 Simulated results

With the methods described in previous sections, we find the following results:

Statistic	Value	95% CI
Emergency patient waiting time	3.0798 min	[3.0669, 3.0927]
Outpatient waiting time	2.7976 min	[2.7696, 2.8256]
Total fraction of patients waiting outside	0.0029	[0.0028, 0.0030]
Scanner utilization in office hours	0.7313	[0.7192, 0.7433]
Scanner utilization outside office hours	0.3549	[0.3513, 0.3584]
Access time outpatients	0.7576 days	[0.7549, 0.7603]
Fraction inpatients called in office hours scanned outside office hours	0.0254	[0.0246, 0.0263]

Table 3: Realized values of the statistics of interest, for a warm-up period of $D = 2 \cdot 10^7$ min and a total running time of 10^8 min.

6 Conclusion

The analytical capacity analysis showed that the two CT scanners currently used by the department are sufficient. During office hours, the expected utilization is around 73%, while outside office hours it is expected to be around 33%.
The results of our simulation show that:

- Emergency patients had to wait approximately 3.08 minutes on average.
- Outpatients had to wait around 2.80 minutes on average after arriving at the CT department.
- Outpatients were able to get an appointment on average around 0.76 days after calling.
- 0.29% of patients could not immediately find an unoccupied chair after arriving in the waiting room.
- Around 73% of the scanners capacity was used during office hours
- Around 36% of the scanners capacity was used outside office hours, a little higher than was expected.
- Approximately 2.5% of inpatients who arrived during office hours could not be scanned before 16:00.

Overall, the results of our simulation indicate that the current situation in the CT department (the amount of scanners and chairs, and the policy for patient scheduling) is sufficient to handle the expected demand while maintaining low waiting times and a very small chance of having to wait outside the waiting room. We argue that no changes in policy or amount of scanners and chairs are required for the current demand.